

Oleh Marushchak

Software Developer

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🏠 Wrocław, PL
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Profile

Motivated and detail-oriented software developer with a strong foundation in algorithms, system programming, and machine learning. Experienced in building performant C/C++ and Python applications, from low-level multithreaded simulations to high-level ML models. Enthusiastic about solving real-world problems through clean, efficient code and continuous learning. Particularly interested in artificial intelligence, embedded systems, and high-performance computing.

Skills

- **Programming Languages:** C/C++, CUDA C/C++, Python, SQL, Bash
- **Machine Learning & Computing:** PyTorch, TensorFlow, NumPy, CuPy
- **Tools & Infrastructure:** Git, Linux, Docker, Kubernetes, SFML, NVIDIA Nsight (nsys, ncu), NVIDIA Omniverse (USD Composer, Isaac Sim, Isaac Lab)
- **Core Domains:** Multithreaded and Low-Level Programming, Machine Learning, High-Performance Computing, Data Structures, Algorithms, Physics, Mathematics

Professional Experience

CUDA C/C++ Developer Nokia

Wrocław, PL

12.2025 – Present

- Designed and implemented GPU-accelerated Layer 1 (L1) 5G algorithms using CUDA and C++, optimizing high-performance computing tasks.
- Profiled and optimized existing C++ codebases using NVIDIA Nsight tools (nsys, ncu), applying low-level programming techniques to reduce latency and improve memory utilization.
- Participated in a collaborative project between Nokia and NVIDIA focusing on advanced telecommunications solutions.

Team Project Member – "Neurocar" Neurosoft (KPZ)

Wrocław, PL

02.2026 – Present

- Developing software solutions for the "Neurocar" project in collaboration with Neurosoft as part of the academic Team Project Course (KPZ PWr).
- Collaborating within a student team using agile methodologies to design, implement, and integrate project components.
- Implemented and evaluated advanced deep learning models for sequence modeling and time-series analysis, including LSTM Autoencoders, TimesNet, and iTransformer.

Projects

Ant Colony Optimization: Sequential vs Multithreaded vs GPU

[\(GitHub\)](#)

Implemented the Ant Colony Optimization (ACO) algorithm in C++ to solve the Travelling Salesman Problem (TSP), comparing different execution strategies across CPU and GPU architectures.

- Developed three ACO variants: sequential, multithreaded (CPU), and GPU-accelerated.
- Benchmarked performance and analyzed execution time on large graph datasets.
- Utilized CUDA and C++ multithreading to demonstrate efficiency gains over the baseline.
- Integrated the SFML library for visualizing algorithm progress in real time.

Clustering of Cocktail Dataset using K-means

[\(GitHub\)](#)

Built a clustering model to analyze and group similar cocktail recipes based on their ingredients and characteristics.

- Applied K-means algorithm in Python to discover structure in beverage data.
- Designed a system to recommend similar cocktails based on a selected input drink.
- Preprocessed and normalized real-world data for clustering accuracy and stability.
- Delivered an intuitive proof-of-concept with visual and textual results.

Parallel and Distributed Computing (In Development)

[\(GitHub\)](#)

Developing a suite of high-performance computing algorithms focused on parallel processing and distributed system architectures.

- Implementing and benchmarking various parallel algorithms using C++, AVX2, MPI, and CUDA to explore performance scaling.
- Designing distributed data processing solutions aimed at minimizing network overhead and optimizing memory access patterns.
- Resolving complex synchronization issues, managing concurrent resources, and ensuring thread safety across multiple nodes.

Arduino-Based Gamepad Controller with Python Integration

[\(GitHub\)](#)

Designed and built a custom joystick-based gamepad using Arduino and Python to simulate keyboard and mouse input via serial communication.

- Collected analog and digital input from joysticks and buttons on an Arduino board.
- Used Python with 'pyserial' and 'pyautogui' to translate hardware signals into PC actions.
- Enabled real-time emulation of keyboard presses, mouse movement, and scrolling.
- Implemented debounce handling, configurable input thresholds, and double-click support.

Education

IT Automation Systems

[Wrocław University of Science and Technology](#)

Wrocław, PL

2023 – Present

Relevant Courses: Automation, Robotics, Machine Learning, Data Science, Data Structures, Databases, Programming, Electronics, Low-Level Programming

Certification: Cisco Networking Essentials (2025)

Languages

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|--|----------------------------|
| • English: Pre-Advanced (B2) | • Ukrainian: Native |
| • Polish: Advanced (C1) | |
| • German: Pre-Intermediate (B1) | • Russian: Native |

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